SPHERE MISSION CONTROL

Spacesuit Thermal Shielding Kit

TEACHER MANUAL AND ASSESSMENT GUIDE

TECHNICAL UNIVERSITY OF MUNICH • SPHERE PROJECT 2026 • INSTRUCTOR EDITION

INSTRUCTOR GUIDE & ANSWER KEY

SPACESUIT THERMAL SHIELDING KIT

INSTRUCTOR MANUAL, SAFETY GUIDANCE AND ASSESSMENT KEY

Scope: The activity is intended for introductory physics teaching. Instructors should adapt the risk assessment, supervision ratio and assessment weighting to local institutional requirements.

1. LEARNING OUTCOMES & CORE MATHEMATICAL THEORY

On completion of this laboratory module, students should be able to:

- Distinguish conduction, convection and radiation mechanisms within the capsule insulation system.
- Apply Newton's law of cooling,

  $$T(t) = T_{\text{env}} + (T_0 - T_{\text{env}})e^{-kt},$$

  and interpret the fitted cooling constant k (min^{-1}).
- Acquire and plot physical temperature observations (T_{obs}) at 60-second intervals ($t = 0$ to 15 min).
- Calculate temperature residuals ($|\Delta T| = |T_{\text{obs}} - T_{\text{model}}|$) and Mean Absolute Error ($\text{MAE} = \frac{1}{n}\sum_{i=1}^{n} |\Delta T_i|$), evaluating model limitations and experimental uncertainty.

1.2 INSTRUCTOR BOARDWORK KEYFRAMES & EQUATIONS

Use these core mathematical formulations to anchor class explanations during introductory chalkboard briefings:

1. SPECIFIC HEAT CAPACITY EQUATION

Calculates total thermal energy dissipation from internal liquid core to surroundings:

$$Q = m \cdot c \cdot \Delta T$$

Where: m = 0.1 kg (100 mL water mass), c = $4184\ \text{J}/(\text{kg} \cdot {}^\circ\text{C})$ (specific heat capacity of liquid water), and ΔT = temperature change (${}^\circ\text{C}$). This quantifies thermal energy lost over time.

2. NEWTON'S LAW OF COOLING (FIRST-ORDER LUMPED MODEL)

Models the transient rate of heat loss as proportional to temperature difference:

$$T(t) = T_{\text{env}} + (T_0 - T_{\text{env}}) \cdot e^{-kt}$$

Where: T_0 = initial water temperature (80.0°C nominal), T_{env} = environmental bath temperature (0.0°C nominal ice-water slurry), k = insulation decay constant (min^{-1}), and t = time (min).

2. 60-MINUTE LESSON ROADMAP

TIMELINE	LESSON PHASE	WHAT YOU & YOUR STUDENTS ARE DOING
$0\text{–}10$ min	1. Theory and concept check	Review the three heat-transfer modes, the water-bath boundary condition ($T_{\text{env}} = 0.0^\circ\text{C}$) and the limits of the spaceflight analogy. Complete the preliminary concept check.
$10\text{–}20$ min	2. Capsule Assembly	Assign teams their insulation setup (A: bare, B: bubble wrap, C: reflective film, D: multilayer assembly). Demonstrate probe placement and the pre-immersion leak test.
$20\text{–}35$ min	3. Fifteen-minute test	Add 100 mL of water at 80.0°C to each capsule. Measure the bath temperature, immerse the capsules, start the timer and record temperatures every 60 seconds.
$35\text{–}48$ min	4. Analysis	Plot measured (T_{obs}) and predicted (T_{model}) temperatures, calculate residuals and MAE ($\text{MAE} = \frac{1}{n}\sum \lvert\Delta T_i\rvert$), and compare the error with instrument accuracy ($\pm 0.5^\circ\text{C}$).
$48\text{–}60$ min	5. Technical discussion	Compare the four configurations and discuss material mechanisms, model assumptions, uncertainty and design trade-offs.

3. LAB SETUP, SAFETY PROTOCOLS & COTS LOGISTICS

Thermodynamic experiments in school and university environments introduce manageable risks. Complete and document the following controls before and during the session:

3.1 CRITICAL CLASSROOM SAFETY CHECKLIST

- **Hot-water control:** Do not use boiling water (100°C). Prepare water at 80.0°C **maximum** and use heat-resistant gloves, eye protection and a stable pouring area.
- **Bath condition:** Prepare an ice-water slurry and measure its temperature. Do not assume an exact 0.0°C boundary without verifying with a calibrated digital probe.
- **Leak Prevention:** Check that students seal the probe opening using the approved lab method and complete a leak test before immersion.

- **Probe Placement:** Check that each probe hangs in the center of the liquid and isn't pressed against the glass jar walls.

3.2 HAZARD RISK ASSESSMENT MATRIX

HAZARD DESCRIPTION	RISK LEVEL	MITIGATION MANDATE
Boiling Water Scalding Using boiling $100^{\circ}\mathrm{C}$ water directly in student environments.	HIGH	Enforce a strict $80^{\circ}\mathrm{C}$ **upper limit**. Provide thermal kettles pre-programmed to $80^{\circ}\mathrm{C}$ or use standard thermometers to lock temperatures before filling glass test capsules.
Ice-Bath Spillages Slips/trips from condensation, melted ice, or water spills around electricity.	MEDIUM	Establish a clear physical separation between ice-water reservoirs and any tablet or laptop charting stations. Provide absorbent microfiber cloths.
Capsule Leakage & Thermal Shock Glass jar thermal shock risk if filled with extreme temperatures (> $90^{\circ}\mathrm{C}$).	LOW	Never pour boiling water directly into glass jars. Allow water to settle to $80^{\circ}\mathrm{C}$, seal lid and probe opening using approved lab method, then leak-test before immersion.

☼ SUMMER & HIGH-AMBIENT TEMPERATURE PROTOCOL ($\sim 30^{\circ}\mathrm{C}$)

- **Reduce Target Water Temp:** Safety scale water down to $55^{\circ}\mathrm{C}$–$60^{\circ}\mathrm{C}$ to neutralize scalding risks on hot days. The software simulator supports any initial value.
- **Ice Melt Compensation:** Instruct students to measure actual bath temperature and input it as T_{env} in Mission Control if ice melts above $0.0^{\circ}\mathrm{C}$.
- **Ice-Free Testing Option:** Drop capsules into room-temperature tap water ($\sim 25^{\circ}\mathrm{C}$–$30^{\circ}\mathrm{C}$) and configure T_{env} accordingly in the software.

3.3 COMMERCIAL OFF-THE-SHELF (COTS) COMPONENTS & COST ESTIMATE

The test rig uses readily available commercial off-the-shelf components to limit cost and simplify replacement:

COMPONENT / MATERIAL	DESCRIPTION & SPECIFICATION	ESTIMATED BUDGET
Glass Test Capsules	4x cylindrical glass jars (100 mL internal water volume)	Confirm supplier quote
Digital Probes	4-pack waterproof digital probe thermometers ($\pm 0.5°C$ accuracy)	~16 €
Suit Insulation Pack	Cotton wool, bubble wrap, reflective Mylar emergency foil sheets	~8 €
Cold Testing Tub	Plastic ice-water immersion basin	~5 €
ESTIMATED TOTAL MATERIALS BUDGET (excl. delivery & consumables)		**~29 €**

Measurement Method & Repeatability: Record thermometer readings manually at 60-second intervals. A consistent probe depth hanging in liquid center and a verified lid seal ensure high repeatability and minimize Mean Absolute Error (MAE) against theoretical predictions.

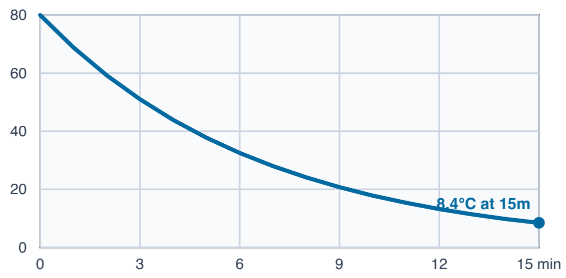
4. REFERENCE-MODEL OUTPUT

The table lists outputs from the assumed coefficients at $t = 15$ min, with $T_0 = 80.0°C$ and $T_{env} = 0.0°C$. These are model values (T_{model}), not acceptance criteria for measured results.

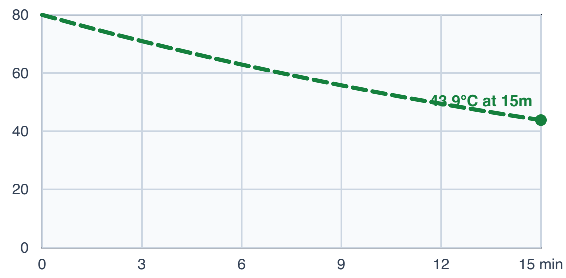
INSULATION TYPE	ASSUMED COOLING CONSTANT (k, min^{-1})	MODEL TEMP AT 15 MIN	QUALITATIVE INTERPRETATION
A. Bare Control	$k \approx 0.150\ \text{min}^{-1}$	$\approx 8.4°C$	Rapid heat loss through unshielded conduction into water and internal liquid circulation.
B. Bubble Wrap	$k \approx 0.040\ \text{min}^{-1}$	$\approx 43.9°C$	Sealed air cells reduce fluid motion and slow heat transfer.
C. Mylar Foil	$k \approx 0.121\ \text{min}^{-1}$	$\approx 13.0°C$	Reflective film is highly sensitive to conductive contact and leakage in the water-bath setup.
D. Multilayer assembly	$k \approx 0.015\ \text{min}^{-1}$	$\approx 63.9°C$	The combined layers increase effective thermal resistance through trapped air, reduced fluid motion and lower radiative exchange.

INDIVIDUAL PREDICTION CURVES ($T_0 = 80.0°\mathrm{C}, T_{\mathrm{env}} = 0.0°\mathrm{C}$)

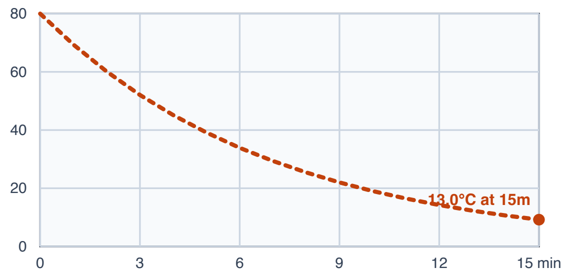
A. BARE CONTROL ($k \approx 0.150\ \mathrm{min}^{-1}$)



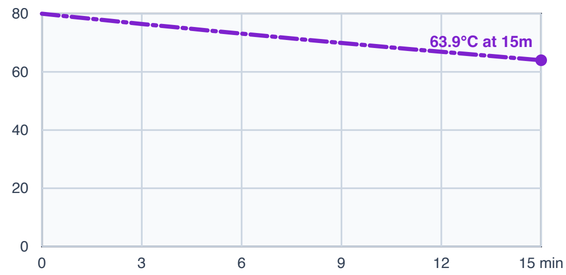
B. BUBBLE WRAP ($k \approx 0.040\ \mathrm{min}^{-1}$)



C. MYLAR FOIL ($k \approx 0.121\ \mathrm{min}^{-1}$)



D. MULTILAYER ASSEMBLY ($k \approx 0.015\ \mathrm{min}^{-1}$)



Custom mode: Mission Control also provides an adjustable k value for inquiry work. It is not shown as a fixed reference curve because its prediction changes with the instructor's selected value. **Optional extension:** A cotton-only configuration may be added as a fifth physical trial when time permits.

5. OFFICIAL SOLUTIONS & TEACHER ANSWER KEY

This solution key covers all ten evaluation questions aligned across the Student Task Sheet (Inquiries 1–10) and the online Mission Control Question Sheet (Questions 1–10). Nine multiple-choice questions are auto-scored on the website; Question 4 requires instructor review and is excluded from the automatic percentage. **Online answer key:** Q1: B, Q2: B, Q3: A, Q4: Instructor Review, Q5: B, Q6: A, Q7: A, Q8: A, Q9: B, Q10: A.

Question 1: Radiant Barriers vs. Direct Conduction (Inquiry 1 / Online Q1 [Key: B])

Sample answer: Reflective film reduces radiative heat exchange, but has negligible thickness and provides limited resistance to conduction when in direct contact with the capsule and ice water. A low-conductivity spacer (cotton or trapped air) is required to block conductive heat flow.

Marking guidance: Award credit for distinguishing radiative reflection from conductive resistance.

Question 2: Understanding the Decay Constant (k) (Inquiry 2 / Online Q2 [Key: B])

Sample answer: Within Newton's Law of Cooling ($T(t) = T_{\text{env}} + (T_0 - T_{\text{env}})e^{-kt}$), a smaller fitted decay constant k (min^{-1}) signifies a slower exponential approach to ambient temperature, indicating superior thermal insulation performance.

Marking guidance: Award credit for connecting smaller k to slower cooling rate and better insulation.

Question 3: Why Multi-Layer Insulation (MLI) Works (Inquiry 3 / Online Q3 [Key: A])

Sample answer: The multilayer assembly combines complementary thermal mechanisms: cotton traps air to block conduction; bubble wrap seals air cells to limit fluid convection; and reflective film reduces radiative exchange, while layer interfaces add contact resistance ($k \approx 0.015\ \text{min}^{-1}$).

Marking guidance: Look for mention of trapped air, reduced convection, radiation reduction, and interfacial resistance.

Question 4: Real-World Experimental Factors & Residuals (Inquiry 4 / Online Q4 [Instructor Review])

Sample answer (Accept any two): (1) Thermometer sensor tip touching cold capsule wall instead of water center. (2) Bath water leaking into capsule through an incomplete lid seal. (3) Ice bath warming up above 0.0°C during the 15-minute trial. (4) Water inside the capsule not stirred, causing internal thermal stratification.

Marking guidance: Open-response item. Award full credit for any two plausible physical causes of model deviation.

Question 5: Ice Bath Temperature Drift & Calibration (Inquiry 5 / Online Q5 [Key: B])

Sample answer: Measure the actual ice-water bath temperature (e.g. $T_{\text{env}} = 4.5^\circ\text{C}$) and update the T_{env} input field in Mission Control so the software recalculates the theoretical reference curve for actual classroom conditions.

Marking guidance: Award credit for specifying that T_{env} must be updated in the software.

Question 6: Effect of Capsule Water Volume / Thermal Mass (Inquiry 6 / Online Q6 [Key: A])

Sample answer: Doubling water volume ($100 \text{ mL} \to 200 \text{ mL}$) doubles thermal mass ($Q = m \cdot c_p \cdot \Delta T$). With twice as much heat energy to dissipate across roughly equal surface area, the capsule cools **more slowly**, and the fitted cooling constant k **decreases**.

Marking guidance: Award credit for noting that cooling is slower and fitted k decreases.

Question 7: Spacesuit Design Trade-Offs & Vacuum Mechanics (Online Q7 [Key: A])

Sample answer: Adding unlimited insulation layers increases suit mass, bulk, and stiffness, severely restricting astronaut joint mobility. Additionally, astronaut metabolic heat must be rejected to prevent dangerous heat buildup inside the suit.

Marking guidance: Accept joint flexibility/mobility restrictions, suit mass/bulk, or metabolic heat dissipation issues.

Question 8: Specific Heat Capacity & Fluid Response (Online Q8 [Key: A])

Sample answer: Replacing water ($c_p \approx 4184 \text{ J}/(\text{kg} \cdot \text{K})$) with a liquid of lower specific heat capacity means the liquid stores less thermal energy per degree. Under equal heat transfer conditions, the lower-capacity liquid exhibits a faster temperature decrease.

Marking guidance: Look for explanation that lower heat capacity results in faster temperature drop.

Question 9: Asymptotic Boundary Limits ($t \to \infty$) (Online Q9 [Key: B])

Sample answer: According to Newton's Law of Cooling ($T(t) = T_{\text{env}} + (T_0 - T_{\text{env}})e^{-kt}$), as time approaches infinity ($t \to \infty$), the exponential term $e^{-kt} \to 0$, so capsule temperature $T(t)$ approaches the environmental bath temperature T_{env} asymptotically.

Marking guidance: Award credit for identifying T_{env} as the asymptotic limit.

Question 10: Telemetry Anomaly Diagnostics & MAE Reduction (Online Q10 [Key: A])

Sample answer: A rapid temperature drop faster than predicted indicates cold water leakage into the capsule or vigorous fluid motion enhancing forced convection. To reduce Mean Absolute Error (MAE), maintain consistent probe placement near liquid center, seal lids thoroughly, and log temperatures at exact 60-second intervals ($\text{MAE} = \frac{1}{n} \sum_{i=1}^{n} |T_{\text{obs},i} - T_{\text{model},i}|$).

Marking guidance: Award credit for identifying fluid leakage or forced convection, and listing key procedural controls for MAE reduction.

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6. GRADING RUBRIC

CRITERIA	EXCELLENT (90-100%)	GOOD (75-89%)	NEEDS WORK (<75%)
Data Logging	All 15 minutes filled with accurate readings, absolute differences, and helpful notes.	Completed with only 1-2 missed minutes; minor calculation mistakes.	Multiple blank rows; missing calculations.
Graphing	Clean, accurate points plotted with a smooth line; dashed theoretical curve clearly shown.	Points plotted with minor curve scaling inaccuracies.	Points not connected; axes mislabeled or missing lines.
Analysis Questions	Clear, thoughtful answers explaining conduction, convection, radiation, and MLI synergy (Q1-Q8).	Good understanding of heat transfer with minor details missing.	Confuses conduction with radiation; incomplete answers.